Zohaib Ali Arif

Embedded Systems Engineer | Hardware, PCB Design & Firmware

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Professional Summary

Embedded engineer working across both sides of the board – high-speed multilayer PCB design and the Embedded C firmware that runs on it. 3+ years of experience taking designs from blank schematic to production hardware: Marvell Ethernet switches, a 12-layer PCIe/USB 3.0/DDR3 board, 300 W GaN converters, STM32 and PIC firmware, with hands-on bring-up and EMI/EMC debugging. Completed a Master's in Embedded Systems; seeking a full-time embedded hardware and firmware role.

Technical Skills

- **Circuit & PCB Design:** Analog Circuit Design, Digital Circuit Design, Mixed-Signal Design, Schematic Capture, Multilayer PCB Layout, High-Speed Digital Design, Signal Integrity, Power Integrity, EMI/EMC, Differential Routing, DFM, IPC-2221
- **Embedded Firmware:** Embedded C, Interrupts, DMA, Timers, ADC/PWM, Board Bring-Up, JTAG/SWD Debugging, Design Verification & Validation, AES-256, Red/Black Separation
- **Platforms & Interfaces:** STM32, PIC, ESP32, Raspberry Pi, Marvell Switch ASICs, i.MX6; UART, SPI, I2C, I2S, CAN, USB 3.0, Ethernet, PCIe, DDR3, GPS/GSM
- **Power Electronics:** GaN Converters, Totem-Pole PFC, LLC Resonant Converters, Thermal Characterization
- **Programming & Tools:** C, Python, MATLAB/Simulink; Altium Designer, LTspice, Multisim, STM32CubeIDE, MPLAB X, Git
- **Lab & Test Equipment:** Oscilloscope (high-voltage differential probes), Power Analyzer, AC/DC Electronic Load, Programmable AC/DC Sources, Logic Analyzer, DMM, Thermal Camera, Microscope, Soldering & Rework

Professional Experience

Embedded Systems Engineer (Master's Internship) *Feb 2026 – Aug 2026*
Wise Integration, Grenoble, France

- Modified and validated 300 W GaN Totem-Pole PFC and LLC resonant converters for AC/DC supplies at 97% peak efficiency.
- Adapted and validated the digital PFC/LLC control firmware in Embedded C for the reworked boards.
- Designed schematics and 4-layer PCB layouts in Altium Designer for 2 WI-7xx GaN power-stage daughter boards.
- Ran bring-up, electrical testing and root-cause debugging on 3 GaN prototypes and customer boards.
- Validated and characterized efficiency and thermal performance across load and line; documented results in technical reports.

Embedded Design Engineer *Jul 2022 – Sep 2025*
Care Pvt. Ltd., Islamabad, Pakistan

- Owned hardware and firmware for Layer 2 managed switches based on Marvell ASICs (24 x 1G copper + 2 x SFP, 2.5 Gbps fiber and generic SFP modules), implementing VLAN and MAC binding, and shipping 3 revisions to production.
- Laid out a 12-layer PCB carrying PCIe, USB 3.0, DDR3 and Ethernet with the hardware team, maintaining signal and power integrity through controlled impedance, length-matched differential routing and thermal optimization.
- Designed the hardware for a secure portable keyboard on STM32 and ESP32 MCUs, applying AES-256 encryption and logical Red/Black separation for protected data transmission.
- Wrote Embedded C firmware for STM32 and PIC (UART/SPI/I2C, interrupts, timers, DMA), integrating GPS/GSM, eMMC, DAC audio, sensors and motor drivers into 3 products.
- Led the embedded hardware team on the DSCR system through design, production and field troubleshooting, working with field support to deploy patches and specify hardware/firmware upgrades.

Projects

i.MX6Q Quad-Core Single-Board Computer – Custom Linux SBC integrating HDMI, PCIe, USB, Ethernet, I2C, SPI and I2S; co-designed the hardware and supported system integration alongside senior engineers.

Underwater GPS Tracking System – IP68-rated MCU system with Quectel GPS, DAC audio, vibration feedback and sensors; 35% longer battery life via low-power optimization.

High-Speed Microcentrifuge (14,000 RPM) – STM32 BLDC motor control with Hall sensors, PID speed loop, safety interlocks and touch GUI.

Education

International Master in Embedded Systems Security (IMESS) *2025 – 2026*
Grenoble INP – Esisar, Valence, France

B.S. Electrical & Electronics Engineering *2018 – 2022*
COMSATS University, Islamabad, Pakistan

Languages: English – C1 | French – A1, currently taking classes